Supplementary

Table S1 The reason and references of excluded full-texts studies

Author	Year	Exclusion reason	Reference
Bailey *et al.*	2015	HCY_Intervention and Osteoporosis	(1-31)
Baines *et al.*	2007		
Bhupathiraju *et al.*	2007		
Blouin *et al.*	2009		
Bozkurt *et al.*	2009		
Bucciarelli *et al.*	2010		
Cagnacci *et al.*	2008		
Dhonukshe-Rutten *et al.*	2003		
Ebesunun *et al.*	2014		
Enneman *et al.*	2015		
Enneman *et al.*	2014		
Garg *et al.*	2014		
Golbahar *et al.*	2004		
Haroon *et al.*	2012		
Karimi *et al.*	2011		
Kim *et al.*	2013		
Kuyumcu *et al.*	2012		
Li *et al.*	2017		
Liu *et al.*	2016		
Morris *et al.*	2005		
Ouzzif *et al.*	2012		
Rehackova *et al.*	2013		
Rejnmark *et al.*	2008		
Rumbak *et al.*	2012		
Salari *et al.*	2014		
Shahab-Ferdows *et al.*	2012		
Tarakida *et al.*	2011		
Vurucu *et al.*	2009		
Weber *et al.*	2016		
Yamada *et al.*	2011		
Yılmaz *et al.*	2009		
Cashman *et al.*	2005	Gene polymorphoism and homosysteinemia	(32-41)
Cook *et al.*	2014		
Guttormsen *et al.*	1996		
Hong *et al.*	2007		
Lacasana *et al.*	2012		
Li *et al.*	2017		
Pandey *et al.*	2013		
Qin *et al.*	2012		
Saito *et al.*	2009		
Tongboonchoo et al	2013		
Bathum *et al.*	2004	Gene polymorphism and Fracture	(42-47)
Chung *et al.*	2012		
Kim *et al.*	2016		
Shiraki *et al.*	2008		
Villadsen *et al.*	2005		
Yazdanpanah *et al.*	2008		
Herrmann *et al.*	2005	No fracture data regarding HCY	(48-58)
Keser *et al.*	2013		
Komulainen-Ebrahim *et al.*	2017		
Kutílek *et al.*	2012		
Lacroix *et al.*	2008		
Meera *et al.*	2010		
Øyen *et al.*	2015		
Rhew *et al.*	2008		
Swart *et al.*	2016		
Tsuchie *et al.*	2016		
Zhu *et al.*	2016		
Ahmadieh H. *et al.*	2011	Review	(59-72)
Bailey *et al.*	2015		
Behera *et al.*	2017		
Clarke *et al.*	2014		
Fratoni *et al.*	2015		
Herrmann *et al.*	2006		
Herrmann *et al.*	2007		
Herrmann *et al.*	2008		
Hiraoka *et al.*	2017		
Mclean *et al.*	2007		
Nieves *et al.*	2012		
Petramala *et al.*	2009		
Saito *et al.*	2006		
Swart *et al.*	2013		
Colson *et al.*	2015	Meta-analysis	(73-77)
Ruan *et al.*	2015		
van Wijngaarden *et al.*	2013		
Yang *et al.*	2012		
Zhang *et al.*	2014		
Ochi *et al.*	2017	Non-research articles	(78-82)
Raisz *et al.*	2004		
Spence *et al.*	2017		
van Meurs *et al.*	2005		
No authors listed	2005		
Bezsmertnyi	2013	No relevance	(83-86)
Lanzoni *et al.*	2017		
Smulders *et al.*	2013		
Tyagi *et al.*	2011		
van Wijngaarden *et al.*	2011	Study rationale and design	(87)
Luo *et al.*	2017	HCY and survival analysis	(88)
Sato *et al.*	2005	Retracted	(89)

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	Study	Items			NOS	Study Design
		Selection	Comparability	Outcome/Exposure		
Observational study	McLean *et al.* 2004	****	**	***	9	Prospective cohort study
	van Meurs *et al.* 2004	****	**	***	9	Prospective cohort study
	Dhonukshe-Rutten *et al.* 2005	****	**	***	9	Prospective cohort study
	Ravaglia *et al.* 2005	****	**	***	9	Prospective cohort study
	Sato *et al.* 2005	****	**	***	9	Cohort study
	Sato *et al.* 2005[$]	****	**	***	9	Prospective control study
	Gerdhem *et al.* 2007	****	**	***	9	Prospective cohort study
	Gjesdal *et al.* 2007	****	**	***	9	Prospective cohort study
	Périer *et al.* 2007	****	**	***	9	Prospective cohort study
	Sawka *et al.* 2007	****	**	***	9	Prospective cohort study
	Yazdanpanah *et al.* 2007	****	**	***	9	Prospective cohort study
	McLean *et al.* 2008	****	**	***	9	Prospective cohort study
	LeBoff *et al.* 2009	****	**	***	9	Case-control study
	Zhu *et al.* 2009	****	**	***	9	Cohort study
	Shiraki *et al.* 2011	****	**	***	9	Prospective cohort study
	El Maghraoui *et al.* 2012	****	**	**	8	Prospective cohort study
	Enneman *et al.* 2012	****	**	***	9	Population-based cohort study
	Kuroda *et al.* 2013	****	**	***	9	Cross-sectional cohort study
	Lewerin *et al.* 2014	****	**	***	9	Prospective cohort study
	Li *et al.* 2014	****	**	**	8	Cross-sectional cohort study
	Urano *et al.* 2014	****	**	***	9	Prospective cohort study
	Torbergsen *et al.* 2015	****	**	**	8	Case-control study

	Study	Items				Jadad	Study Design
		Generation of random sequence	Allocation concealment	Blinding	Withdrawal and dropout		
Experimental study	Armitage *et al.* 2010	**	**	/	*	5	RCT
	Gommans *et al.* 2013	**	**	/	*	5	RCT
	Wijngaarden *et al.* 2014	**	**	/	*	5	RCT
	Lopez *et al.* 2017	**	**	/	*	5	RCT
	NOS, Newcastle-Ottawa scale; RCT, randomized controlled trial						

Figure S1 Quality assessment of included studies. [$], article retracted; *, 1 point; **, 2 points; ***, 3 points; ****, 4 points.

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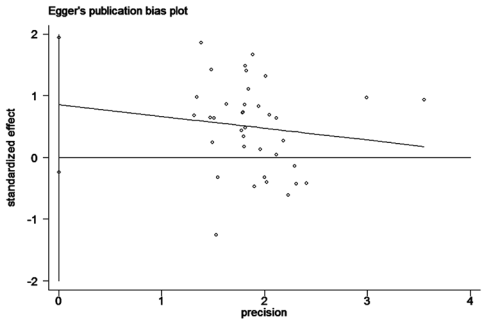


Figure S2 Egger's publicaiton bias plot.

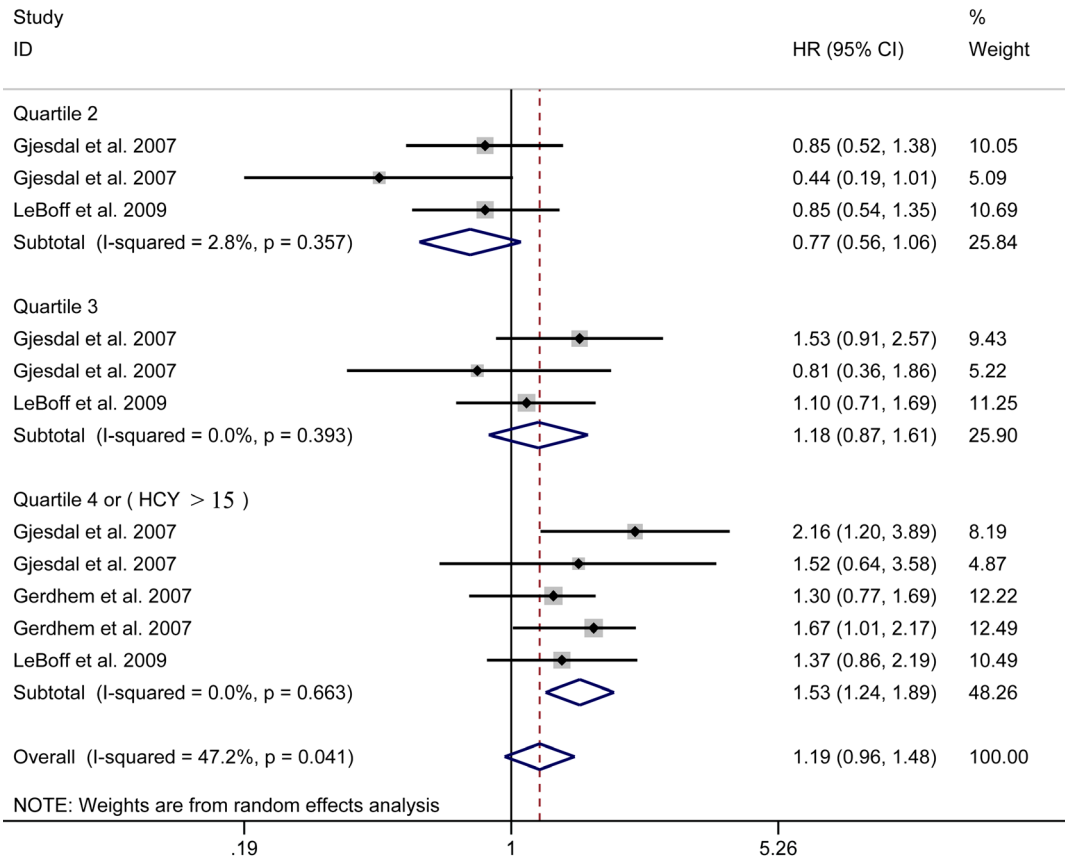


Figure S3 Subgroup analysis for hip fractures demonstrated that the results did not change either with overall data, or in the second, third and highest quartile.

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